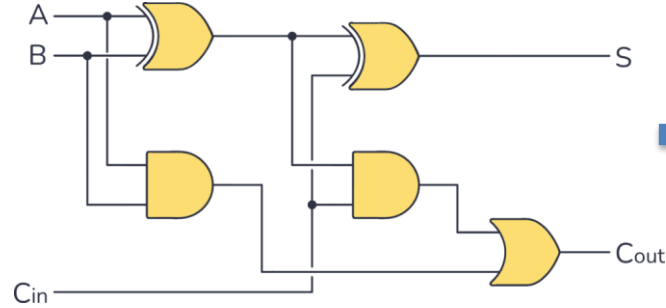
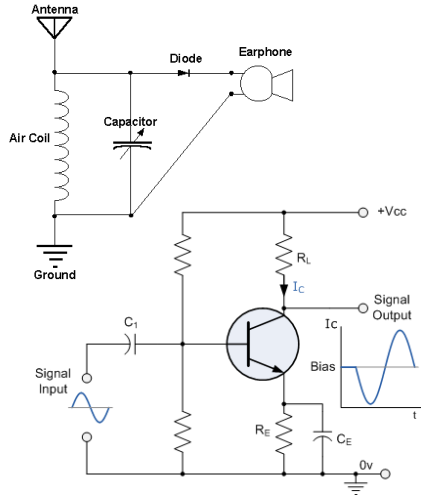


How Python and AI helped keep me from getting thrown out of a saloon

James Abel

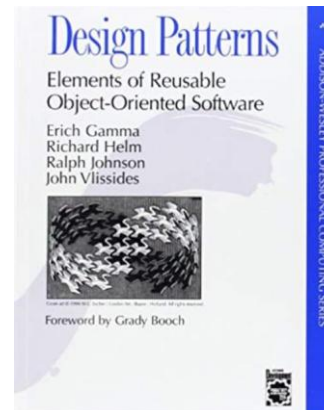
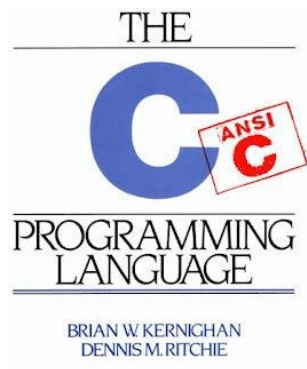
8/27/26

My Journey ...



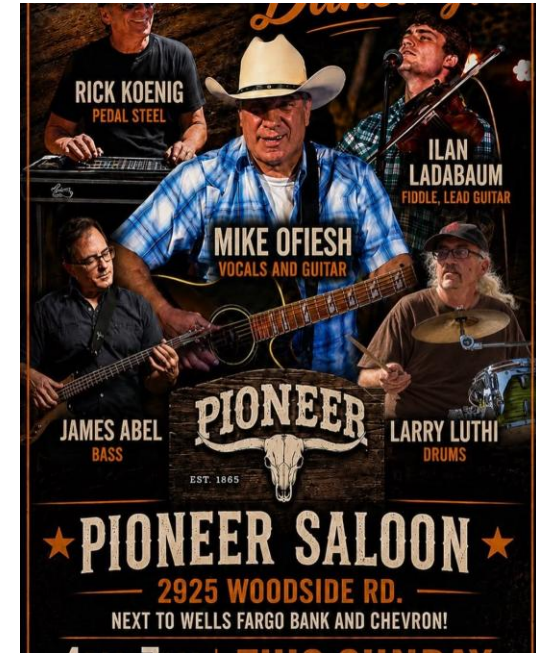
AUTOSTART ROM LISTING

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00EB	235	*****
00EC	236	*****
00ED	237	*****
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00EF	239	*****
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00F9	249	*****
00FA	250	*****
00FB	251	*****
00FC	252	*****
00FD	253	*****
00FE	254	*****
00FF	255	*****



Set List (song, key)

People are crazy	F	Luchenbach ,Texas -G	Theme from Dukes-E
Cruise	A	Fishin in the dark D	Two dozen roses G
Be My Baby tonight	A	Down on the farm A	Chasin that neon r G
Some Girls Do.	D	Shoulda been a cowboy -G	Cant You see - D
Small Town Saturday	G	Family tradition - E	Big city E
Daddy's Money.	A	Boot Scoot Boogie -E	On the Road Again- E
Chatahoochie	C	Friends in low places A	Chicken fried -G
She thinks my tractors	G	Cadillac Ranch. E	Margaritaville - D
Brand new man	G	Folsom - E	Something like that - F
Write this down	F	Neon Moon- A	Achy breaky heart A
Pickup Man	G	Watermelon Crawl - G	Wagon Wheel - A



- Transposition
- Stem Extraction
- Stem Solo and Mute

JBL BandBox Solo 18-watt 1 x 2.25-inch Practice Amplifier

18W, 1 x 2.25" Modeling Guitar Amp with Tone/Amp Models, Guitar/Mic In, AI Stem Separation, Bluetooth, Tuner, Rechargeable Battery, and Mobile App Support

★★★★☆ 7 reviews | [Write your review](#) | [Share](#) | Item ID: BandBoxSolo



\$249.95

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FREE Shipping

✓ In Stock! **PLU**

Ad



PRO

Advanced AI tools for creators

\$16.66

Per month, paid annually

Pro Plan

MOST POPULAR

Access to our best models and editing tools.

\$8 /month

Songstem Demo



Songstem

Playlist: MikeOfiesh

Songs (checked are processed)

Select all Select none

Song	Pitch (1/2-steps)
☐ Billy Currington — People Are Crazy (re-recorded)	0 st ^ v
☐ Florida Georgia Line — Cruise (re-recorded)	-1 st ^ v
☐ John Michael Montgomery — Be My Baby Tonight (re-recorded)	0 st ^ v
☐ Sawyer Brown — Some Girls Do (re-recorded)	0 st ^ v
☐ Hal Ketchum — Small Town Saturday Night (re-recorded)	-1 st ^ v
☐ Ricochet — Daddy's Money (re-recorded)	0 st ^ v
☐ Alan Jackson — Chattahoochee (re-recorded)	0 st ^ v
☐ Kenny Chesney — She Thinks My Tractor's Sexy (re-recorded)	-3 st ^ v
☐ Brooks & Dunn — Brand New Man (re-recorded)	0 st ^ v
☒ George Strait — Write This Down (re-recorded)	-1 st ^ v
☐ Joe Diffie — Pickup Man (re-recorded)	0 st ^ v
☐ Waylon Jennings — Luckenbach, Texas (Back to the Basics of Love) [feat. Willie Nelson] (re-recorded)	0 st ^ v
☐ Nitty Gritty Dirt Band — Fishin' In the Dark (re-recorded)	0 st ^ v

Separation

Isolate stem:

Bass

Muted-mix levels (per stem):

Vocals 100%

Bass 0%

Drums 100%

Guitar 100%

Piano 100%

Other 100%

C:\Users\JamesAbel\projects\songstem\temp

Output...

Run

Re-record playlist → WAV (loopback)

Create pitch-shifted WAVs

Progress

ffmpeg found: C:\Users\JamesAbel\AppData\Local\Microsoft\WinGet\Links\ffmpeg.EXE

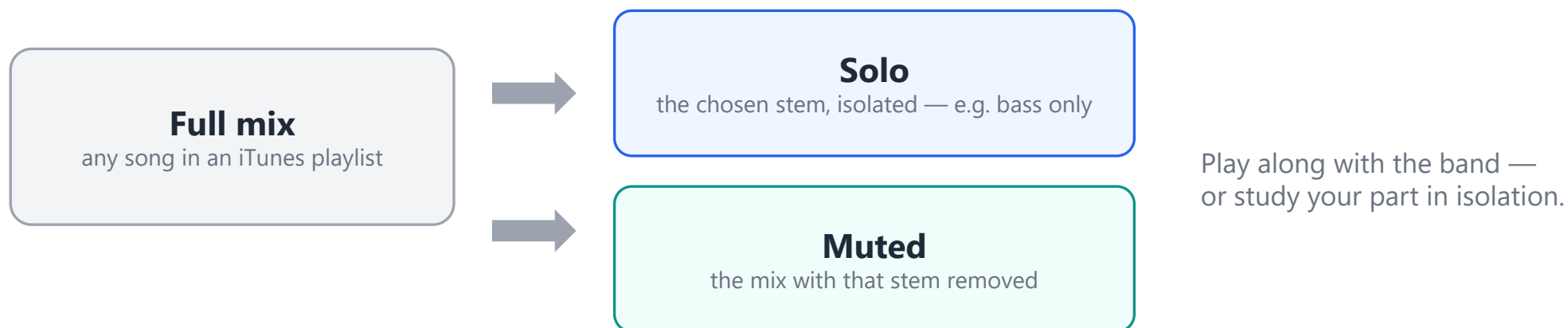
Player

Play Pause Stop Vol



Songstem

Instrument removal for Apple Music on Windows — batch stem separation for music practice



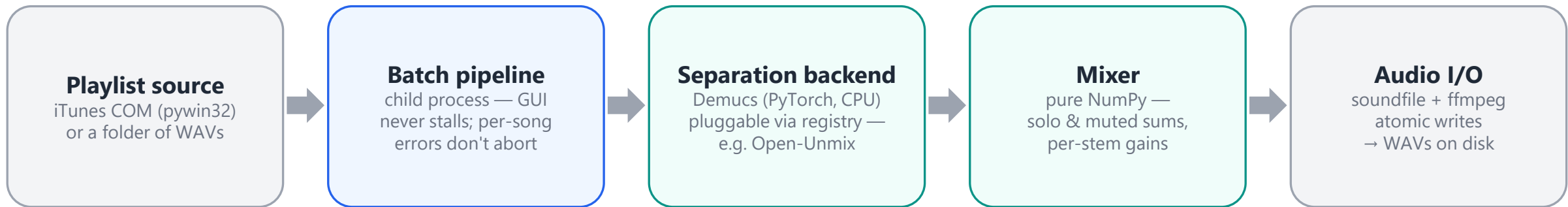
Windows · Python · optimized for practice and cover-song rehearsal — quality over latency (batch, not live)

Architecture



Layered pipeline — every heavy dependency sits behind a swappable seam

GUI — PySide6 (Qt 6): playlist picker · stem & level controls · progress/Stop · built-in player (Qt Multimedia) · UI state in SQLite (pref)



Design rule

Heavy/platform imports (Demucs, Qt, COM, soundfile) stay local behind ABCs with in-memory fakes — the core is importable and tested headless.

Analysis (best-effort, never fails a job)

librosa: key (Krumhansl–Schmuckler) · tempo · note patterns
sync'd lyrics (network) → Markdown cheat sheet

Underlying technologies

All Python — Windows x86 desktop, Python 3.14

Separation / ML

- Demucs v4 — hybrid transformer source separation
- PyTorch, CPU inference (GPU-ready via config)
- Backend registry → swappable engines

Desktop UI

- PySide6 (Qt 6) widgets
- Qt Multimedia output player
- QThread workers + child process for heavy work

Audio processing

- NumPy mixing · soundfile (libsndfile) I/O
- ffmpeg for AAC/ALAC decode (auto-installed via winget)
- librosa — key/tempo/note analysis, pitch shifting

Windows integration

- pywin32 — iTunes/Apple Music COM automation
- sounddevice + VB-Audio Virtual Cable loopback capture
- Runs on regular CPython (not free-threaded 3.14t)

Persistence & state

- pref — SQLite-backed UI state
- Remembers playlist, checked songs, pitch, window
- Atomic file writes for crash-safe outputs

Quality & tooling

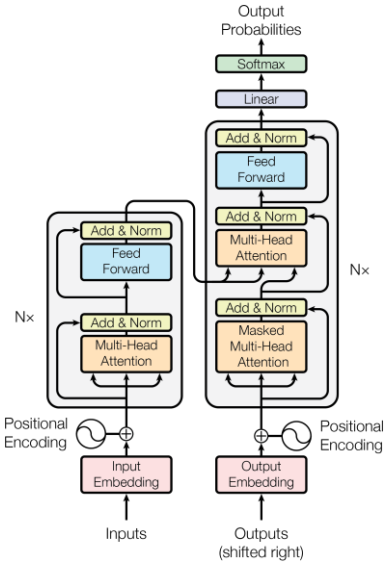
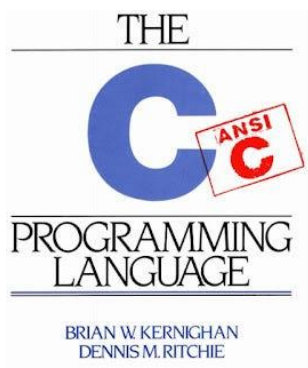
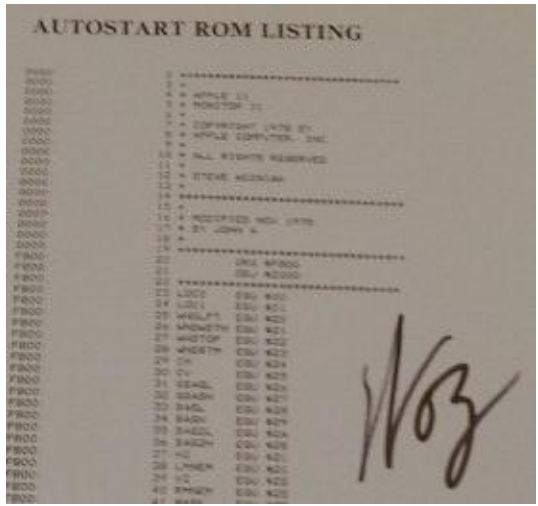
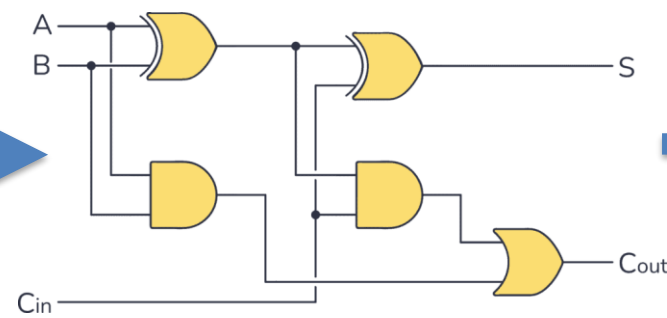
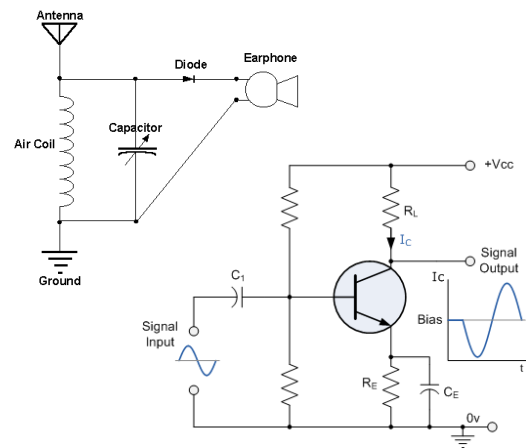
- pytest suite over the dependency-light core
- Fakes for iTunes, separation, recording, analysis
- ruff linting · hatchling packaging

Every dependency is plain PyPI — no services, no accounts; everything runs and stays on the local machine.

Development

- All done in ~1 day with Claude Code
 - Simple interactive prompts
 - Iterative
- I'm responsible for functionality – via use
- I wrote no Python
- Claude Code added regression tests

My Journey ...



Transformer model architecture

Summary

- With AI, less of the work is now coding per-se.
- The **Responsibility** of the delivered product hasn't changed.
- **Specifications** (Prompts) is a skill.
 - What technologies should be applied to the end product?
- **Tool selection** is important (what model, security, cloud vs. local, cost, etc.).
 - What AI techniques exist and should be applied?
 - What are Transformers, Generative Pretrained Transformer (GPT), RAG, MCP, Agents, etc.
- **Tests** are as important as ever (if not more so).
- The responsible party must still be able to ensure things are **fixed** when they go wrong.
- <https://github.com/jamesabel/songstem>

THE
PELICAN INN

*10 pacific way,
Muir beach, California 94965*

Mon Sept 7, 2026

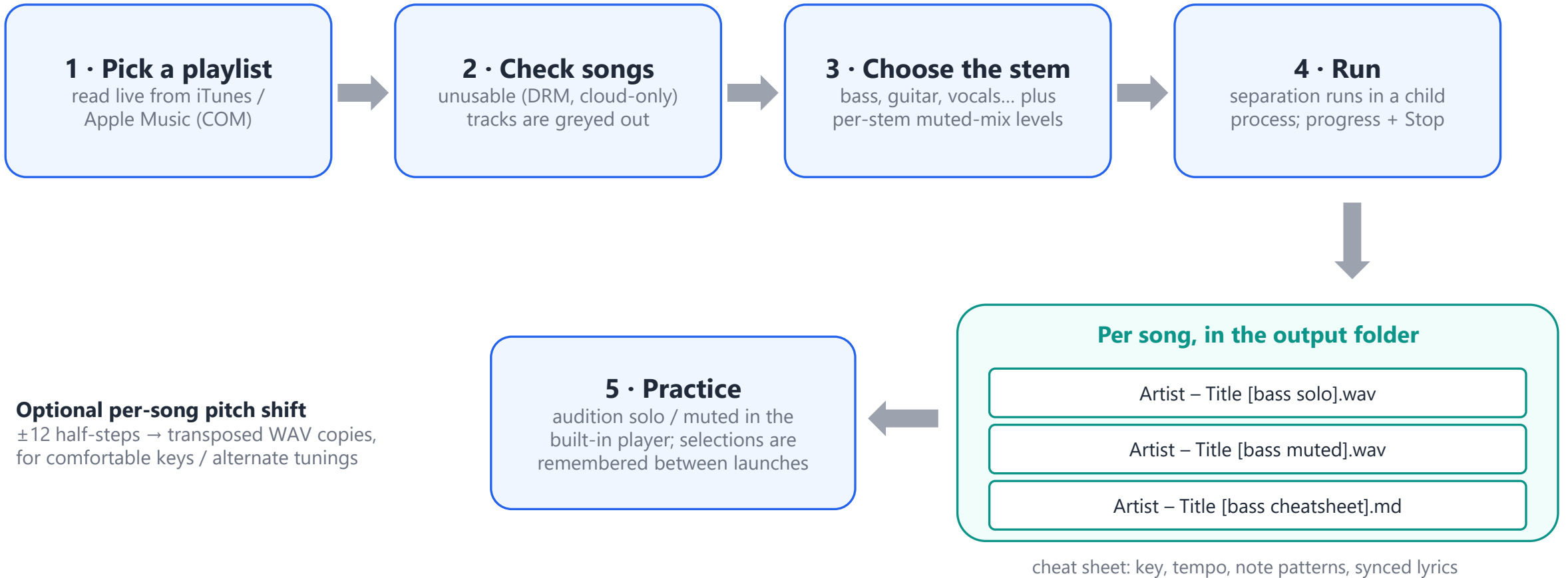


BACKUP

Usage model



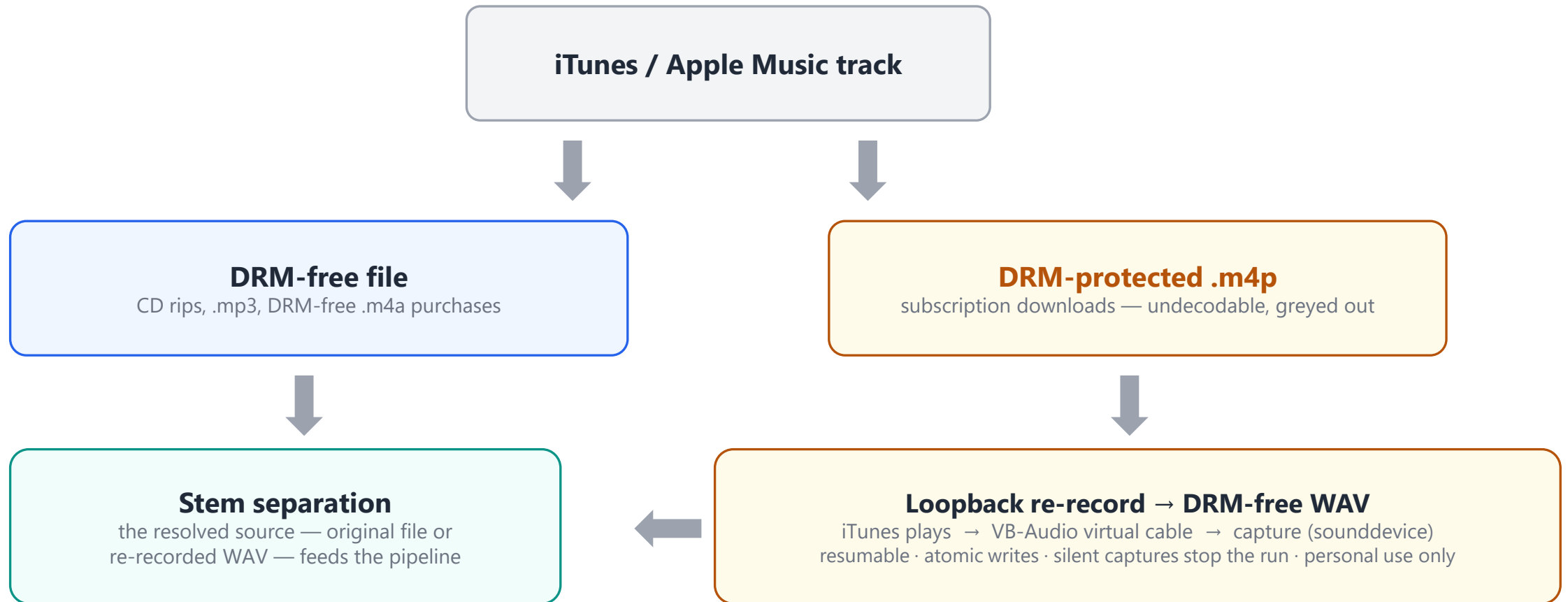
One button after setup — the batch runs in the background



Audio sources and the DRM boundary



Separation needs decodable audio — Apple's FairPlay DRM cannot be decrypted



Re-recording captures playback in real time (a playlist takes about its runtime); it records audio output — it does not remove DRM.

Testing

- Owner-Generated or AI?
 - Avoid “tests are boring – I’ll let AI do it”
- Owner
 - For critical code and/or delivering to others, owner’s tests should specify functionality (TDD, Spec Driven Tests, etc.).
 - For non-critical and/or hyper-personalized code (code is tested during use), user-written tests may not be necessary.
- AI
 - AI can create Regression Tests once functionality is defined. Enables refactoring, performance improvements, framework migration.
- Use Code Coverage (including branch coverage) to guide test efforts.
 - 80% Code Coverage → 20% “not covered”.
- The expertise is in deciding what tests to create and by whom.

Finish up with my tech over time, with AI added.